

- 2.3. If cascaded, stages having only *second-order* nonlinearity can yield a finite IP_3 . For example, consider the cascade identical common-source stages shown in Fig. 2.75.

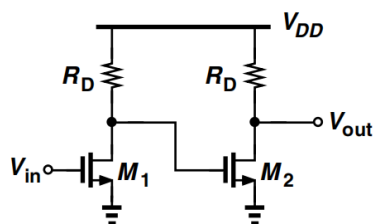


Figure 2.75 Cascade of CS stages.

If each transistor operates in saturation and follows the ideal square-law behavior, determine the IP_3 of the cascade.

- 2.5. Consider the scenario shown in Fig. 2.76, where $\omega_3 - \omega_2 = \omega_2 - \omega_1$ and the band-pass filter provides an attenuation of 17 dB at ω_2 and 37 dB at ω_3 .

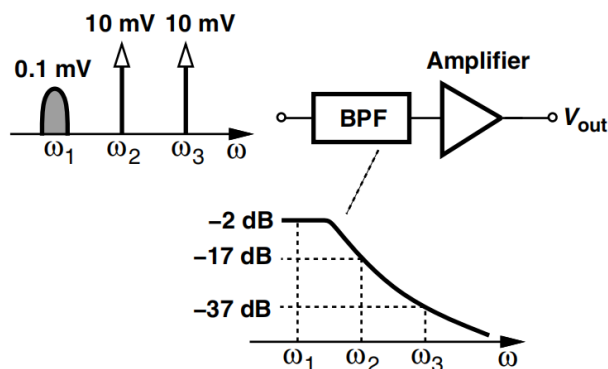


Figure 2.76 Cascade of BPF and amplifier.

- Compute the IIP_3 of the amplifier such that the intermodulation product falling at ω_1 is 20 dB below the desired signal.
 - Suppose an amplifier with a voltage gain of 10 dB and $IIP_3 = 500 \text{ mV}_p$ precedes the band-pass filter. Calculate the IIP_3 of the overall chain. (Neglect second-order nonlinearities.)
- 2.21. Determine the noise figure of each circuit in Fig. 2.79 with respect to a source impedance R_S . Neglect channel-length modulation and body effect.

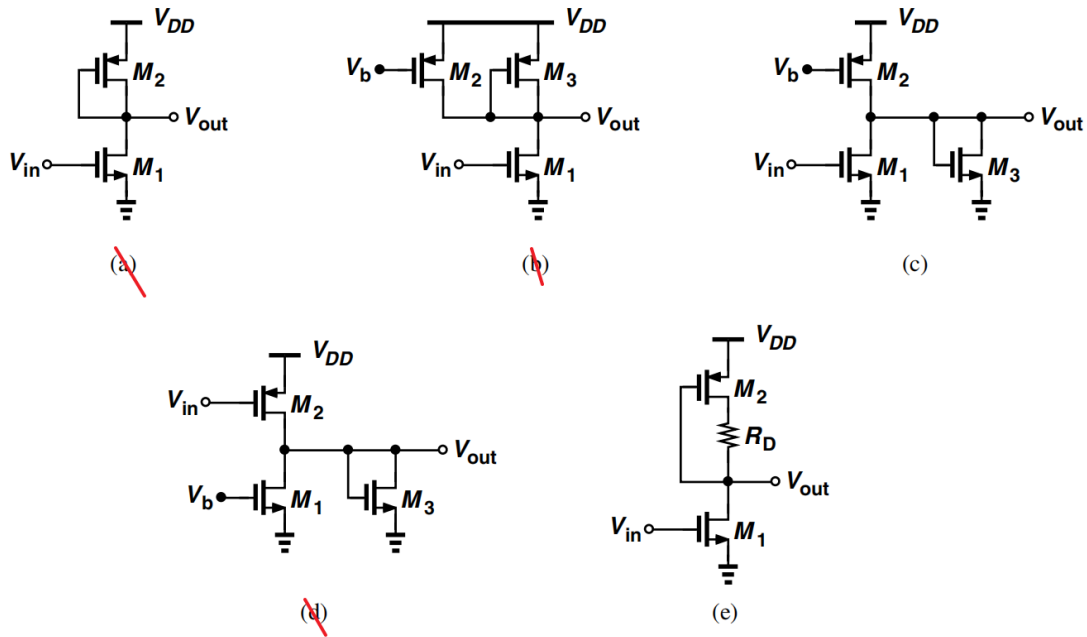


Figure 2.79 CS stages for NF calculation.

2.23. Determine the noise figure of each circuit in Fig. 2.81 with respect to a source impedance R_S . Neglect channel-length modulation and body effect.

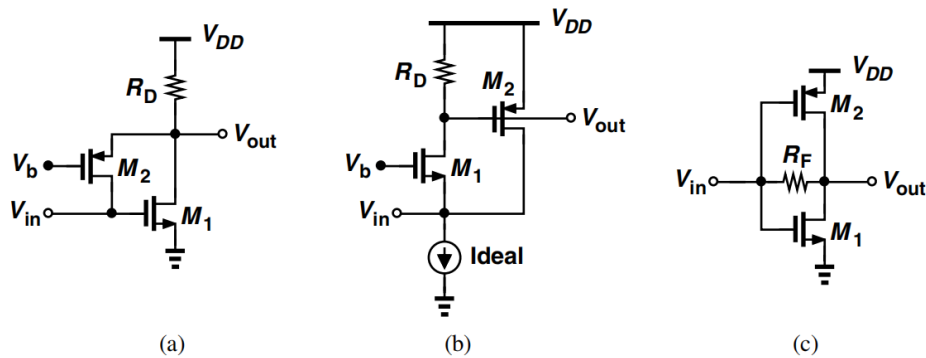


Figure 2.81 Stages for NF calculation.

(5) Consider the following receiver architecture with the individual specs shown for each block (assume all the blocks are matched to 50 ohms). Determine the total noise figure $(NF)_{dB}$. Also determine the total sensitivity. The specs for the IF filter and BPF3 are considered together as Block 5. If the total IIP3 of the receiver is -15 dBm find the SFDR of the receiver.

